



Bilimni
baholash
agentligi



BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA
Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA

FANIDAN

TEST TOPSHIRIQLARI KITOB

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **Shohruh Matematika jamoasi**

Telegram bot: [@bm_rashtest_bot](https://t.me/bm_rashtest_bot)

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

- yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.
- ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'inishda yozing.
- moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO‘PLAMI!
SHOHRUH MOCK-3. 01.10.2026

1. $a^2 + 5 = 2b - 6c$ ifodada a , b va c lar natural sonlar. Quyidagilardan qaysi biri doim to‘g‘ri?
- A) a juft son
B) a toq son
C) b va c toq sonlar
D) b va c juft sonlar
2. $10! + 7!$ soni nechta nol bilan tugaydi?
- A) 3
B) 1
C) 2
D) 5
3. O‘quvchi uyidan maktabgacha 3 km/h tezlikda yurib, darsga 1 min kechikdi. Boshqa safar u 4 km/h tezlikda yurib, dars boshlanishidan 3 min oldin yetib keldi. Agar u har safar uydan bir xil vaqtda chiqsa, dars boshlanishiga aynan o‘z vaqtida kelishi uchun keyingi safar qanday tezlikda yurishi kerak?
- A) 3, 2
B) 1, 6
C) 3
D) 1, 5
4. Umumiy bahosi 225 dinor bo‘lgan ikki xil qimmatbaho mo‘ynali teri xalqaro bozorda 40% foydasi bilan sotildi. Agar birinchi xil teridan 25%, ikkinchisidan 50% foyda qilingan bo‘lsa, har bir terini bahosi necha dinor?
- A) 100; 125
B) 90; 135
C) 200; 25
D) 80; 145

5. Hisoblang:

$$\left(\left(\frac{1}{2} \right)^{-2} - 5 \cdot 2^{-2} + \left(\frac{2}{3} \right)^{-2} \right) : (3^0 + 2^{-2}) .$$

- A) 4
B) 5
C) 3
D) 6

6. Ifodaning qiymatini toping:

$$\frac{\sqrt{(\sqrt{11}-4)^2} + \sqrt{12+\sqrt{44}}}{10}.$$

- A) $\frac{1}{2}$
- B) $\frac{\sqrt{3}}{2}$
- C) $\frac{\sqrt{5}}{2}$
- D) 3

7. Hisoblang:

$$\frac{\sqrt{5}-\sqrt{3}+\sqrt{15}-3}{\sqrt{15}-\sqrt{5}+\sqrt{3}-3}.$$

- A) $2-\sqrt{3}$
- B) $2+\sqrt{3}$
- C) $\sqrt{3}-2$
- D) $\sqrt{3}-1$

8. Arifmetik progressiyaning ayirmasi noldan farqli bo'lib, to'rtinchidan o'n to'rtinchisigacha bo'lgan hadlar yig'indisi 77 ga teng. Uning 7 ga teng bo'lgan hadining nomerini toping.

- A) 8
- B) 6
- C) 9
- D) 10

9. Cheksiz kamayuvchi geometrik progressiyaning hadlari yig'indisi 6 ga, hadlari kvadratlarining yig'indisi esa 12 ga teng. Uning hadlari kublarining yig'indisini toping.

- A) 27
- B) 31
- C) $\frac{206}{7}$
- D) $\frac{216}{7}$

10. Soddashtiring:

$$|2x-6|+|x+1| \quad (-1 \leq x \leq 3).$$

- A) $3x-5$
- B) $5-x$
- C) $x-7$
- D) $7-x$

11. Ifodani soddalashtiring:

$$\left(\frac{a^6 + b^6}{a^4 - b^4} + \frac{a^2 b^4 - a^4 b^2}{a^4 + b^4 - 2a^2 b^2} \right) : (a - b) - b .$$

- A) $a - b$
- B) b
- C) a
- D) 1

12. $\log_{600} 900$ ni a va b orqali ifodalang, bu yerda $a = \log_5 2$, $b = \log_2 3$.

- A) $\frac{1 + a + ab}{2 + 3a + ab}$
- B) $\frac{2(1 + a + ab)}{2 + 3a + ab}$
- C) $\frac{1 + a + ab}{2a + 2 + ab}$
- D) $\frac{2(1 + a + ab)}{2a + 2 + ab}$

13. Tenglamani yeching:

$$2 \sin^3 x - 3 \sin x = 2 \cos^3 x + \cos x .$$

- A) $\frac{\pi}{4} + \pi k, k \in \mathbb{Z}$
- B) $\frac{\pi}{4} + 2\pi k, k \in \mathbb{Z}$
- C) $-\frac{\pi}{4} + \pi k, k \in \mathbb{Z}$
- D) $-\frac{\pi}{4} + 2\pi k, k \in \mathbb{Z}$

14. Tengsizlikni yeching:

$$\frac{15^x - 3^{x+1} - 5^{x+1} + 15}{-x^2 + 2x} \geq 0 .$$

- A) $(0; 2)$
- B) $(0; \log_3 5) \cup (\log_3 5; 2)$
- C) $[\log_5 3; \log_3 5]$
- D) $(0; \log_5 3] \cup [\log_3 5; 2)$

15. x va y haqiqiy sonlar uchun

$$\begin{cases} x^2 + 8xy = 60 \\ y^2 - 3xy = -15 \end{cases}$$

tengliklar o'rinli bo'lsa, $x \cdot y$ ko'paytmasini toping.

- A) 3
- B) 4
- C) 5
- D) 6

16. Tengsizlikni qanoatlantiruvchi eng kichik butun sonni toping.:

$$\sqrt{x^2 - 3x + 2} \leq x - 1 .$$

- A) 1
- B) 2
- C) 3
- D) 0

17. Tenglama nechta haqiqiy ildizga ega:

$$\log_{x+1}(x^2 + x - 6)^2 = 4 .$$

- A) 2 ta
- B) 3 ta
- C) 1 ta
- D) 4 ta

18. Tenglamaning haqiqiy ildizlari yig'indisini toping (Agar u bitta bo'lsa o'sha ildizini toping):

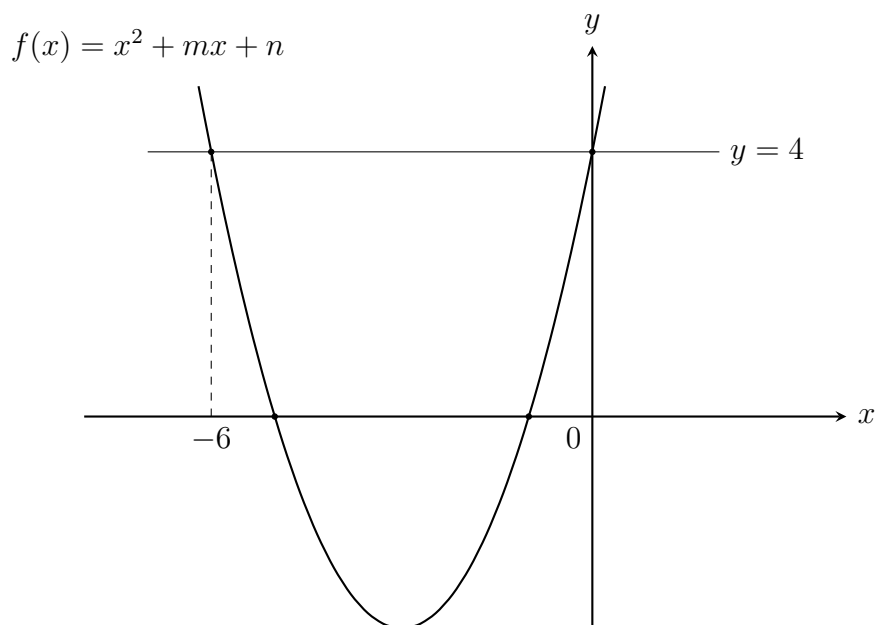
$$-\frac{|x|}{x} - x = \frac{x^2}{2} + 1 .$$

- A) 1
- B) -1
- C) 3
- D) -2

19. $1912 \cdot x^2 + 114 \cdot x - 2026 = 0$ tenglamaning ildizlaridan biri quyidagilardan qaysi biri?

- A) $-\frac{1}{2026}$
- B) 1
- C) $-\frac{1}{1912}$
- D) $-\frac{1912}{2026}$

20. Quyidagi shaklda $f(x) = x^2 + m \cdot x + n$ parabola bilan $y = 4$ to'g'ri chizig'i $x = 0$ va $x = -6$ absissali nuqtalarda kesishadi.



Bunga ko'ra, $f(-3)$ qiymatini toping.

- A) -5
 B) -2
 C) -3
 D) -4
21. $f(x) = x - 4$, $g^{-1}(f(x)) = -x + 7$ bo'lsa, $g(2x + 1)$ ni toping.
- A) $-2x + 2$
 B) $-2x - 2$
 C) $2x - 4$
 D) $-2x + 5$

22. Agar $f(x) = |\log_2(\sin x + \cos x) - 1|$ bo'lsa, $f'\left(\frac{\pi}{4}\right)$ ni toping.

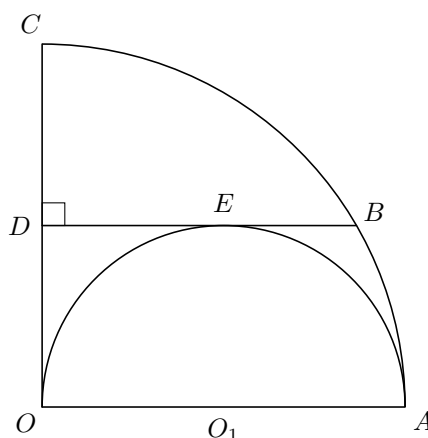
- A) 0
 B) $\frac{1}{\ln 2}$
 C) 1
 D) $-\frac{1}{\ln 2}$

23. $f(x) = ax^2 + bx + c$ funksiya uchun $f'(1) = 4$, $f(0) + f'(0) = 5$ va $\int_0^1 f(x) dx = \frac{7}{3}$ bo'lsa, $a + b - c$ ni toping.

- A) 3
 B) $4, 5$
 C) 2

D) 3, 5

24. Yuqoridagi shaklda O_1 markazli yarim aylana O markazli chorak aylanaga A nuqtada va BD kesmaga esa E nuqtada urinadi, bunda $BD \perp OC$. BC yoy necha gradusligini toping?



- A) 40°
 B) 45°
 C) 50°
 D) 60°

25. Uchburchak ichidan ikkita nuqta shunday olinganki, ularning biridan uchburchak tomonlarigacha masofalar 1 cm, 3 cm va 15 cm bo'lib, ikkinchisidan uchburchak tomonlarigacha (xuddi shu ketma-ketlikda) masofalar 4 cm, 5 cm va 11 cm ga teng. Shu uchburchakka ichki chizilgan aylana radiusini toping.

- A) 5 cm
 B) 6 cm
 C) 7 cm
 D) 8 cm

26. Tomonlari 14 va 5 ga teng parallelogramning katta tomoniga yopishgan burchaklari bissektrisalari bu tomonga parallel tomonni uch qismga ajratadi. Bu qismlardan kichigini toping.

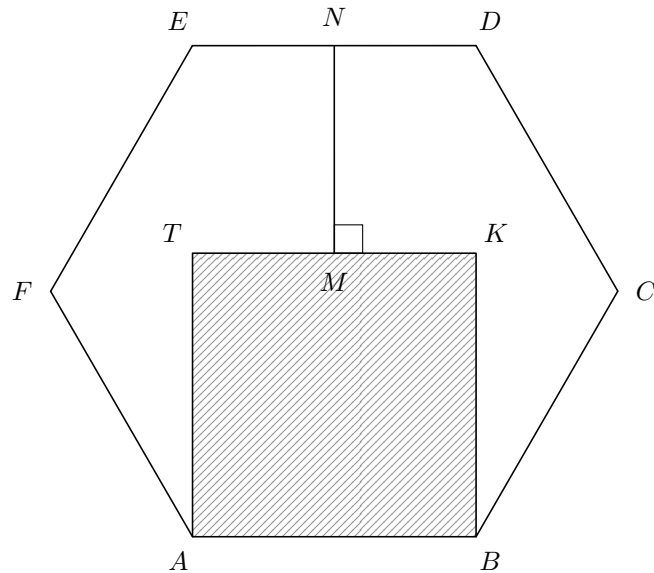
- A) 2
 B) 3
 C) 4
 D) 5

27. Burchagi 60° ga teng, katta asosi 10 ga teng bo'lgan teng yonli trapetsiyaga aylana ichki chizilgan. Trapetsiya kichik asosining uchi va aylana markazi orasidagi masofani toping.

- A) $\frac{4\sqrt{2}}{3}$
 B) $\frac{3\sqrt{3}}{2}$
 C) $3\frac{2}{5}$

D) $3\frac{1}{3}$

28. $ABCDEF$ muntazam oltiburchak va $ABKT$ kvadratda $NM \perp TK$ va $NM = (\sqrt{3} - 1)$ cm bo'lsa, $ABKT$ kvadrat perimetrini toping.

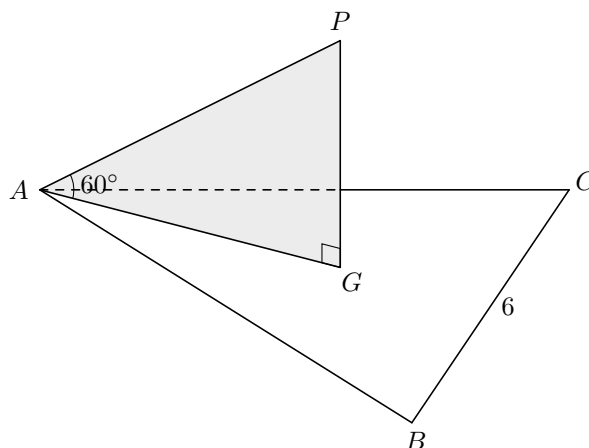


- A) 3
B) 2
C) 4
D) 6

29. $|\vec{a}| = 3$ va $|\vec{b}| = 4$, a va b vektorlar orasidagi burchak 60° ga teng bo'lsa, $\vec{c} = 3\vec{a} + 2\vec{b}$ vektor uzunligini toping.

- A) $\sqrt{215}$
B) $\sqrt{214}$
C) $\sqrt{216}$
D) $\sqrt{217}$

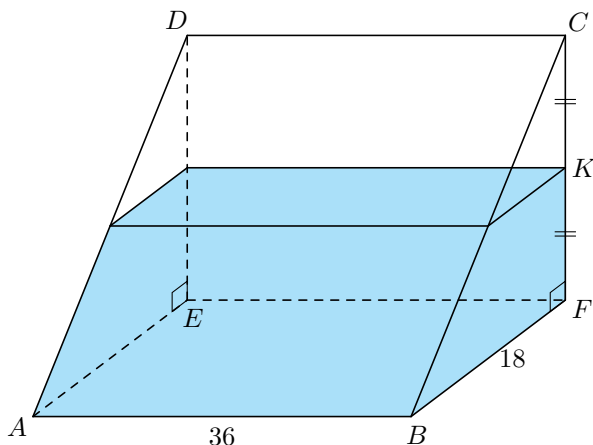
30. ABC teng tomonli uchburchak, G — og'irlik markazi, $BC = 6$ cm, $\angle GAP = 60^\circ$. Shaklda GP kesmasi ABC uchburchagi joylashgan tekislikka perpendikulyar bo'lsa, GP uzunligini toping.



- A) $2\sqrt{3}$
 B) 4
 C) 5
 D) 6
31. $A = \{x \mid |x - 3| \leq 39, x = 2k, k \in \mathbb{Z}\}$ va $B = \{x \mid |x - 9| \leq 36, x = 3k, k \in \mathbb{Z}\}$ to'plamlar berilgan. $n(A \cup B)$ ning qiymatini toping.
- A) 42
 B) 47
 C) 51
 D) 53
32. Kitob javonida kimyo faniga oid 3 ta, matematika faniga oid 4 ta va fizika faniga oid 3 ta kitob mavjud. Tarkibida har bir fanga oid kamida bitta kitob bo'ladigan kitoblar to'plamini necha xil usulda tanlash mumkin?
- A) 315
 B) 525
 C) 735
 D) 945

Topshiriqlar (33–35) va javob variant (A–F) larni o‘zaro moslashtiring.

Quyidagi shaklda to‘g‘ri burchakli uchburchakli to‘g‘ri prizma shaklidagi yopiq idish ichida ma’lum miqdorda suv bor. Idishning $ABFE$ yuzasi yerda bo‘lganda suv balandligi FC qirrasining yarmiga teng bo‘ladi. $AB = 36$ dm, $BF = 18$ dm va $CF = 24$ dm bo‘lsa,



Javoblar.

A) 5832

B) 2916

C) 27

D) 2

E) $\frac{9}{5}$

F) 270

33. Idishdagi ho‘llagan yuza(suv tegib turgan) ning ho‘llanmagan yuzaga nisbatini toping.

34. Idish ichida qancha miqdorda suv bor (litr)

35. Idish AED uchburchakli yuzi yerga tegib turadigan qilib vertikal holatga keltirilganda, idishdagi suvning balandligi necha cm ga teng bo‘ladi?

36. Quyidagi tenglama berilgan:

$$a|x + 1| + (1 - a)|x - 1| + 2 = 0 .$$

a) a ning eng kichik natural qiymatini topingki, yuqoridagi tenglama aynan ikkita turli ildizga ega bo‘lsin.

Javob: _____

b) a ning shunday barcha qiymatlarini topingki, bunda yuqoridagi tenglama ildizga ega bo‘lmasin.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko‘chirib yozing.

37. Tenglamani yeching:

$$2 \operatorname{tg} x + \operatorname{tg} 2x = \operatorname{tg} 4x.$$

a) Tenglamaning $(0; 2\pi)$ intervalga tegishli ildizlari sonini toping.

Javob: _____

b) Tenglamaning $(0; \pi)$ intervalga tegishli ildizlari yig'indisini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraqasiga ko'chirib yozing.

38. $f(x)$ funksiya uchun $f(x) + x \cdot f(1) + x^2 \cdot f(2) = x + x^2$ tenglik o'rinli.

a) $f(4)$ ni hisoblang.

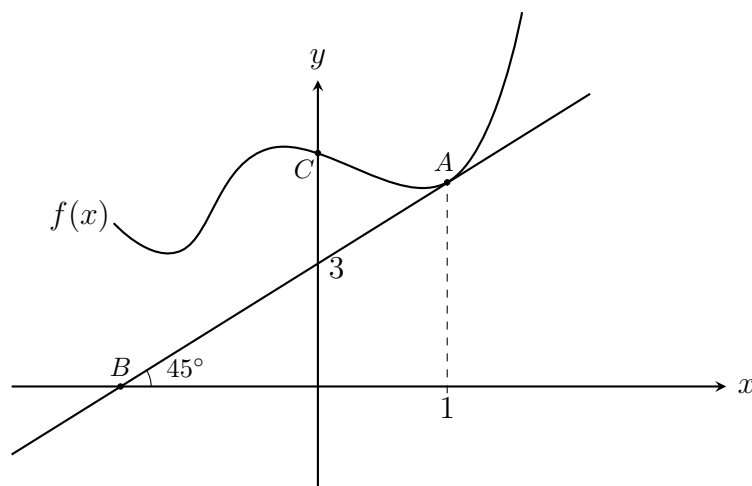
Javob: _____

b) $(6; 8)$ nuqtadan $f(x)$ funksiyagacha bo'lgan eng qisqa masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraqasiga ko'chirib yozing.

39. Quyidagi rasmda $f(x)$ funksiyaning $x = 1$ abssissali nuqtasidagi urinmasi ko'rsatilgan. $h(x) = x \cdot f(x^2)$ bo'lsa,



a) $h'(1)$ ni toping.

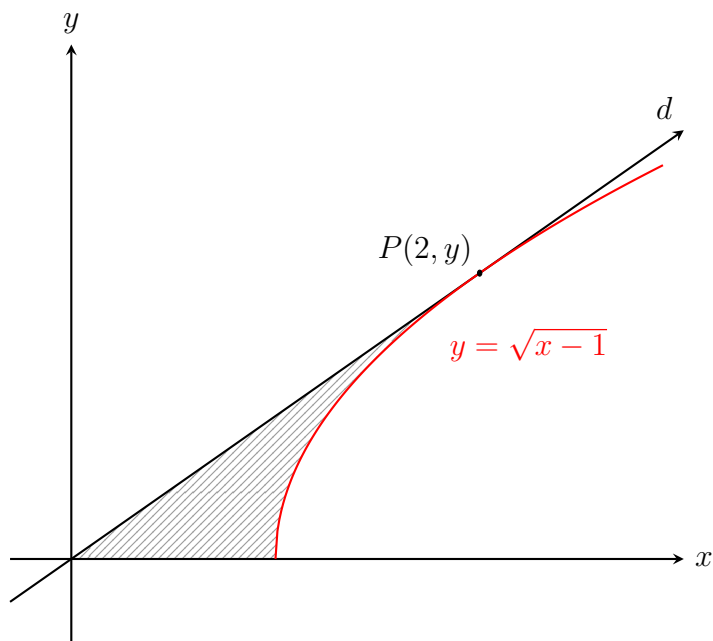
Javob: _____

b) $h(x)$ funksiyaga $x = 1$ nuqtada o'tkazilgan urinma tenglamasini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraqasiga ko'chirib yozing.

40. Quyida $y = \sqrt{x-1}$ funksiyaga $x = 2$ nuqtasiga urinuvchi urinma o'tkazilgan.



a) Shtrixlangan soha yuzini toping.

Javob: _____

b) Shtrixlangan soha yuzini Ox o'qi atrofida 360° ga aylantirishdan hosil bo'lgan jism hajmini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

41. ABC teng yonli o'tkir burchakli uchburchakning yon tomoni 15 ga, yuzi esa 67,5 ga teng. AC asosiga va BC tomoniga o'tkazilgan BE va AH balandliklar O nuqtada kesishadi.

a) BOH uchburchagining yuzini toping.

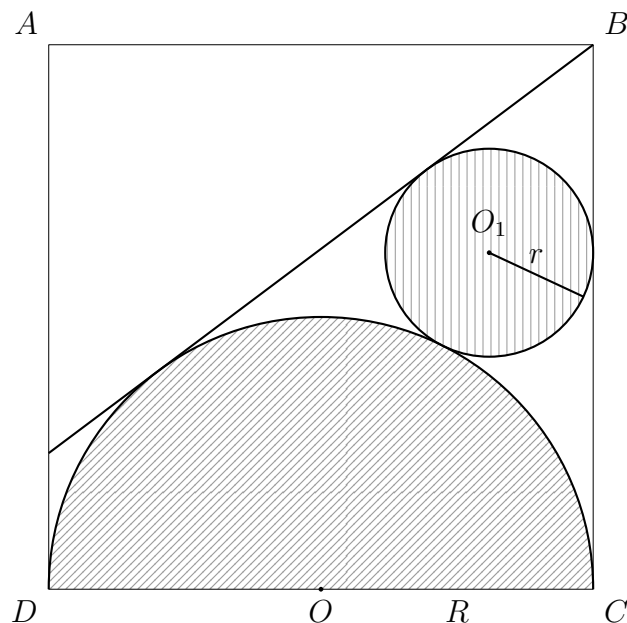
Javob: _____

b) BOH uchburchakning perimetrini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

42. $ABCD$ kvadratga O markazli yarim aylana va O_1 markazli aylana ichki chizilgan.



a) $\frac{R}{r}$ ning qiymatini toping.

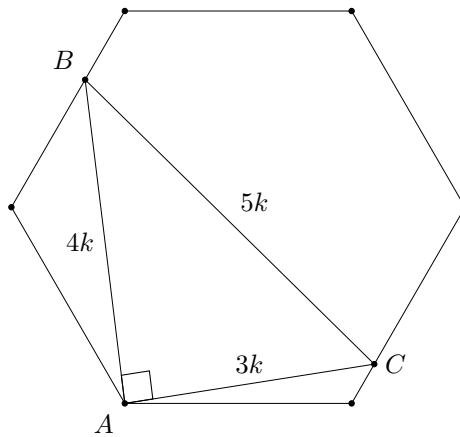
Javob: _____

b) Kvadratning tomoni $3 + \sqrt{5}$ ga teng bo'lsa, kichik aylana yuzasini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

43. Yuzasi 144 sm^2 bo'lgan muntazam oltiburchak berilgan. Unga tomonlari $3k$, $4k$ va $5k$ ga teng bo'lgan to'g'ri burchakli uchburchak ichki chizilgan.



a) $k^2\sqrt{3}$ ning qiymatini toping.

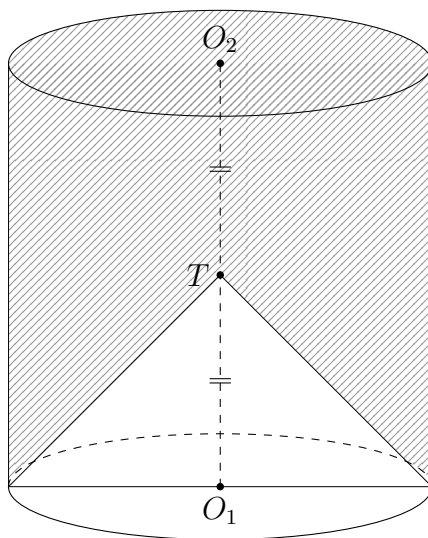
Javob: _____

b) ABC uchburchak yuzini (cm^2) toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

44. Shaklda asoslari ustma-ust tushuvchi to'g'ri konus va to'g'ri silindr berilgan va $O_1T = O_2T$ tenglik o'rinli. Shtrixlangan soha silindrdan konusni chiqarib tashlash orqali hosil qilingan.



- a) Shtrixlangan soha silindrdan konusni chiqarib tashlash orqali hosil qilingan bo'lsa, shtrixlangan soha hajmining konus hajmiga nisbati qanchaga teng?

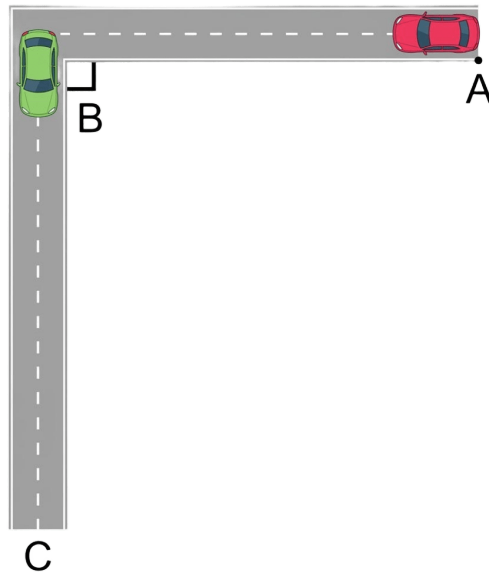
Javob: _____

- b) Konus hajmining silindr hajmiga nisbati qanchaga teng?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

45. Quyidagi rasmda ko'rsatilganidek, A va B nuqtalaridan tezliklari mos ravishda 10 km/soat va 15 km/soat bo'lgan ikkita avtomobil belgilangan yo'nalishlarda bir vaqtda harakat boshlamoqda. $|AB| = 80 \text{ km}$ bo'lsa,



a) Necha soatdan keyin ular orasidagi masofa eng qisqa bo'ladi?

Javob: _____

b) Bu vaqtda birinchi avtomobil qancha masofani bosib o'tgan bo'ladi?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.